

Internship in Physical Organic Chemistry

Week 1 — Part 2: Finding and collecting relevant literature

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Goals:

- Understand what makes a paper relevant *before* you start searching
- Build a search vocabulary from the anchor papers provided by your supervisor
- Collect and organise a working set of 8–12 papers in Zotero
- Select 3–5 papers that will form the basis of your Background chapter

In this part you will build your working literature collection. The objective is not to find every paper that exists on your topic, but to identify a small, coherent set of papers that you can read carefully and use to write your Background chapter.

Scientific literature search has a specific structure: you start from a fixed point, extract vocabulary, search with that vocabulary, screen results systematically, and stop when you have enough material to proceed. Each of these steps is described below.

1 Before you search: the anchor papers

Searching for papers without a reference point is inefficient. Without knowing what a relevant paper looks like, you cannot decide quickly whether a result is useful or not.

Your supervisor has provided you with two anchor papers. These are papers that are known to be relevant and appropriate for your level. Your first task is to read their titles, abstracts, and keywords carefully — not the full text yet.

Task. For each anchor paper, open it and locate the following:

- The **keywords** listed by the authors (usually just below the abstract)
- Two or three **synonyms** used in the abstract for the main concepts
- The **type of paper** it is: review, experimental, computational, or mixed

Write these down. They will become your search terms in Section 3.

This step teaches an important habit: search vocabulary comes from the literature itself, not from your own intuitions. Experts build search terms by reading; they do not guess them.

Illustrative example. Suppose your topic involves a pericyclic reaction. After reading the anchor papers, you might extract terms such as: *retro-Diels-Alder*, *pericyclic reaction*, *ring opening*, *activation barrier*, *concerted mechanism*, *orbital symmetry*. These describe the **chemistry**. Avoid using instrument or method names as your primary search terms — these describe the tool, not the reaction, and will return papers focused on methodology rather than mechanism.

2 Writing your relevance criteria

Before opening Google Scholar, you must write down what a relevant paper looks like. This is done in three sentences, completed before any searching begins.

Task: In your lab notebook or Overleaf document, complete the following three sentences:

1. A relevant paper **must contain**: _____
2. A relevant paper **must not be**: _____
3. I am **not sure** whether I need papers that: _____

The third sentence is the most important. It forces you to notice your uncertainty before you encounter it in the search results.

Papers matching sentence 1 are collected immediately. Papers matching sentence 2 are discarded immediately. Papers that fall into sentence 3 are set aside and examined more carefully in Section 5.

Illustrative example: For a project on gas-phase fragmentation mechanisms:

1. Must contain: experimental fragmentation data with fragment assignments, or a mechanistic discussion of why specific fragments form.
2. Must not be: a paper focused only on sample preparation, chromatographic separation, or biological activity with no structural characterisation.
3. Not sure: papers that study a related but different compound class using the same technique — they may provide useful comparison or context.

3 Searching for papers

You will use Google Scholar as your primary search engine. Open it in your browser and search using the vocabulary you extracted from the anchor papers in Section 1.

Use short, specific queries of two to four keywords. Do not use full sentences. Run several different searches rather than one broad search. Each search should probe a slightly different aspect of your topic.

Task: Run at least four searches using different combinations of the terms you extracted. For each search, open the first five to eight results. Do not read the papers in full. Instead, look at:

- The title and abstract
- The figures, if they are visible in the preview
- The journal and year of publication

Apply your relevance criteria immediately. Import relevant papers to Zotero using the browser connector. Set aside uncertain cases without discarding them.

At the end of this step you should have approximately 8 to 12 papers in your Zotero collection. The exact number is not critical. What matters is that each paper you have collected passed at least sentence 1 of your relevance criteria.

to something in the anchor papers. A paper that introduces an entirely new system or method with no connection to your topic should be set aside for now.

Task. Apply the two-pass screening procedure:

1. For uncertain papers: open the full text, check figures and results section, decide yes or no.
2. For all remaining papers: compare to anchor papers and keep only those with a direct connection to your topic.

At the end of this step you should have 3 to 5 papers. Record their full references in Zotero and make sure each has a PDF attached.

This final selection is the set of papers you will read carefully in Part 3. For each paper you select, fill in a copy of the **Paper Reading Template** (`reading_template.pdf`, provided separately). Complete at minimum Section A before moving on. It is not a permanent decision — you may add or remove papers later as your understanding develops.